# Лабораторная работа

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## Дано

1. $s_1 = \sigma(x_1)$, $s_2 = \sigma(x_2)$
2. $o_{\max} = \max(s_1, s_2)$, $o_{\min} = \min(s_1, s_2)$
3. $y = o_{\max} + o_{\min}$

## Решение

$$1. \frac{\partial E}{\partial o_{\max}} = \frac{\partial E}{\partial y} \cdot \frac{\partial y}{\partial o_{\max}} = \frac{\partial E}{\partial y} \cdot \frac{\partial (o_{\max} + o_{\min})}{\partial o_{\max}} = \frac{\partial E}{\partial y}$$

$$2. \frac{\partial E}{\partial o_{\min}} = \frac{\partial E}{\partial y} \cdot \frac{\partial y}{\partial o_{\min}} = \frac{\partial E}{\partial y} \cdot \frac{\partial (o_{\max} + o_{\min})}{\partial o_{\min}} = \frac{\partial E}{\partial y}$$

$$3. \frac{\partial y}{\partial s_1} = \frac{\partial (o_{\max} + o_{\min})}{\partial s_1} = \frac{\partial (s_1 + s_2)}{\partial s_1} = 1$$

$$4. \frac{\partial y}{\partial s_2} = \frac{\partial (o_{\max} + o_{\min})}{\partial s_2} = \frac{\partial (s_1 + s_2)}{\partial s_1} = 1$$

$$5. \frac{\partial s_1}{\partial x_1} = s_1 \cdot (1 - s_1)$$

**ответ**

$$\frac{\partial E}{\partial x_1} = \frac{\partial E}{\partial y} \cdot \frac{\partial y}{\partial s_1} \cdot \frac{\partial s_1}{\partial x_1} = \frac{\partial E}{\partial y} \cdot (s_1 - s_1^2)$$

$$\frac{\partial E}{\partial x_2} = \frac{\partial E}{\partial y} \cdot \frac{\partial y}{\partial s_2} \cdot \frac{\partial s_2}{\partial x_2} = \frac{\partial E}{\partial y} \cdot (s_2 - s_2^2)$$